

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Di-n-butylamine**  
 Product Description: **Di-n-butylamine**  
 Cat No. : A11671  
 Synonyms N-Butyl-1-butanamine  
 CAS No 111-92-2  
 Molecular Formula C8 H19 N

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                     |
|------------------------------------|---------------------|
| Flammable liquids                  | Category 3 (H226)   |
| Acute oral toxicity                | Category 3 (H301)   |
| Acute dermal toxicity              | Category 3 (H311)   |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H226 - Flammable liquid and vapor

H330 - Fatal if inhaled

H314 - Causes severe skin burns and eye damage

H301 + H311 - Toxic if swallowed or in contact with skin

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - Wear respiratory protection

#### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

EUH071 - Corrosive to the respiratory tract

Lachrymator (substance which increases the flow of tears)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| Di-n-butylamine | 111-92-2 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In

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|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                                 |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

## **Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

## **Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### **Environmental precautions**

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Do not flush into surface water or sanitary sewer system.

## **Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### **Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Corrosives area.

### **Specific End Uses**

Use in laboratories.

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **Control Parameters**

| Component       | European Union | The United Kingdom | Germany                                                                                                                   |
|-----------------|----------------|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Di-n-butylamine |                |                    | TWA: 5 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 29 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>Haut |

### **Exposure Controls**

#### **Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### **Personal protective equipment**

##### **Eye Protection**

Goggles

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

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sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Particulates filter conforming to EN 143 Ammonia and organic ammonia derivatives filter Type K Green conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance** No information available

**Physical State** Liquid

**Odor** Rotten-egg like

**Odor Threshold** No data available

**pH** 11.1

**Melting Point/Range** -62 °C / -79.6 °F

**Softening Point** No data available

**Boiling Point/Range** 159 °C / 318.2 °F

**Flash Point** 39 °C / 102.2 °F

@ 760 mmHg

**Method -** No information available

**Evaporation Rate** No data available

**Flammability (solid,gas)** Not applicable

Liquid

**Explosion Limits** **Lower** 0.6 Vol%

**Upper** 6.8 Vol%

**Vapor Pressure** 2.3 mbar @ 20 °C

**Vapor Density** 4.5

(Air = 1.0)

**Specific Gravity / Density** 0.760

**Bulk Density** Not applicable

Liquid

**Water Solubility** 4.05 g/L (25°C)

**Solubility in other solvents** No information available

### Partition Coefficient (n-octanol/water)

**Component** **log Pow**

Di-n-butylamine 2.1

**Autoignition Temperature** 260 °C / 500 °F

**Decomposition Temperature** No data available

**Viscosity** 0.9 mPa s at 20 °C

**Explosive Properties**

explosive air/vapour mixtures possible

**Oxidizing Properties** No information available

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Molecular Formula C<sub>8</sub> H<sub>19</sub> N  
Molecular Weight 129.24

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous Reactions  
Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Temperatures above 40°C. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Acids. Strong oxidizing agents. Amines. Chlorine. Acid anhydrides. Acid chlorides. Carbon dioxide (CO<sub>2</sub>). halogenated agents.

### Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral Category 3  
Dermal Category 3  
Inhalation Category 2

| Component       | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation      |
|-----------------|--------------------------|-----------------------------|----------------------|
| Di-n-butylamine | LD50 = 189 mg/kg ( Rat ) | LD50 = 768 mg/kg ( Rabbit ) | > 2 mg/L ( Rat ) 1 h |

| Component       | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-----------------|-----------------------|-------------------------|-----------------------------|
| Di-n-butylamine | 220 mg/kg bw          | 300 mg/kg bw            | 1,2 mg/L (vapours)          |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

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**(d) respiratory or skin sensitization;**

**Respiratory**  
**Skin**

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

**(h) STOT-single exposure;**

Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;**

**Target Organs**

Based on available data, the classification criteria are not met

None known.

**(j) aspiration hazard;**

Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component       | Freshwater Fish                                | Water Flea                              | Freshwater Algae                                                                                                                                                                                                                            | Microtox             |
|-----------------|------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Di-n-butylamine | LC50: = 5.5 mg/L, 96h<br>(Oncorhynchus mykiss) | EC50: = 66 mg/L, 48h<br>(Daphnia magna) | EC50: = 19 mg/L, 96h static<br>(Pseudokirchneriella subcapitata)<br>EC50: = 19 mg/L, 96h<br>(Pseudokirchneriella subcapitata)<br>EC50: = 16.4 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: = 1.16 mg/L, 96h<br>(Desmodesmus subspicatus) | EC50 = 196 mg/L 17 h |

**Persistence and degradability**

**Persistence**  
**Degradation in sewage treatment plant**

Expected to be biodegradable  
Persistence is unlikely.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

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|                                  |                |                                      |
|----------------------------------|----------------|--------------------------------------|
| <b>Bioaccumulative potential</b> |                | Bioaccumulation is unlikely          |
| <b>Component</b>                 | <b>log Pow</b> | <b>Bioconcentration factor (BCF)</b> |
| Di-n-butylamine                  | 2.1            | No data available                    |

**Mobility in soil** The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information** Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not empty into drains Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
UN-No UN2248  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name DI-n-BUTYLAMINE

**Road and Rail Transport**  
UN-No UN2248  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name DI-n-BUTYLAMINE

**IATA**  
UN-No UN2248  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name Di-n-BUTYLAMINE

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

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## Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed

| Component       | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Di-n-butylamine | 203-921-8 | X    | X   | X     | X    | X    | X     | X    | KE-04223 |

## National Regulations

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Prepared By

Health, Safety and Environmental Department

### Revision Date

19-May-2025

### Revision Summary

SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

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The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**